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A stand of trees does not a forest make: Tree plantations and forest transitions

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ABSTRACT

Global programs are calling to increase tree cover, including plantations, which supply global pulp and wood demand, energy, food, and carbon markets. Tree plantations that replace native forests, cultivated agriculture, or previously cleared land are essentially commodity crops with global market drivers, and do not provide the same ecosystem services as native forests. Nonetheless, they are counted as "forest" by global programs. We test whether 1) the forest transitions framework (FTF), which typically explains reforestation, adequately describes the socio-economic drivers of plantation establishment and 2) descriptions of the effects of land cover change on ecological processes are obscured when tree plantation and native forest classes are aggregated. We used longitudinal multi-temporal satellite imagery (1985–2001) to map and analyze plantation systems across a 35,853 km^2 area in southern Chile at the plantation frontier. As predicted by the FTF, plantations were established in foothills of predominantly agricultural watersheds rather than in watersheds dominated by native forests or in flat, agriculturally productive areas. Half of the plantations were planted on agricultural or cleared lands that were deforested years ago. Counter to predictions of the FTF, the other half of the plantations replaced native forests. Tree plantations were not associated with rural population loss; instead their establishment was related to the amount of potential usable land. We find that when native forests and tree plantation classes are disaggregated, land in coastal catchments that were converted to tree plantation is related to lower quality nearshore resources; analyses that aggregate plantations with native forests obscure this effect.
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1. Introduction

Relatively new programs such as the Reducing Deforestation and Forest Degradation (REDD+) program of the United Nations Framework Convention on Climate Change (UNFCCC), and the clean development mechanism (CDM) of the Kyoto Protocol call for the increase in tree cover that includes both native forests and plantations (Gullison et al., 2007). Conservation scientists call for landscape change frameworks to direct these efforts (Angelsen and Rudel, 2013) and careful attention to the tree species and planting strategies that comprise these mitigation strategies because maximum rates of carbon sequestration may lead to unanticipated conservation outcomes (Putz and Redford, 2009). Tree plantations have become a major land use globally (Rudel, 2009; Lambin and Meyfroidt, 2010). Since 1989, wood products supplied from native forests have been declining, with plantations filling the gap of roundwood supply (Warman, 2014). According to higher projection estimates by the FAO, tree plantations, excluding palms, will account for ~7% of global forests by 2030, (Penna, 2010). Tree plantations increasingly meet the market demand for global pulp, energy, wood, food, and carbon storage (Sedjo, 1999; Berndes et al., 2003; Gullison et al., 2007; Gutierrez-Velez et al., 2011; Popp et al., 2011). Government subsidies have lead Brazil, Chile, China, India, and Russia to become important tree-plantation regions for wood pulp production (Sedjo, 1999; Barr and Cossalter, 2004; Bull et al., 2006; Zhang and Song, 2006). Since the late 1970′s, the global area of wood-pulp plantations has steadily increased, especially in the southern hemisphere (Sedjo, 1999) and tree plantations have expanded dramatically in Asian countries (Barbier et al., 2010). Pine (*Pinus* spp.) and eucalyptus (*Eucalyptus* spp.) plantations are the most commonly planted trees for wood and pulp (Sedjo, 1999), although people may also plant teak, poplar, acacia, and other trees such as palm and rubber (Gutierrez-Velez et al., 2011). Often these plantations are comprised of trees that are not native to the area

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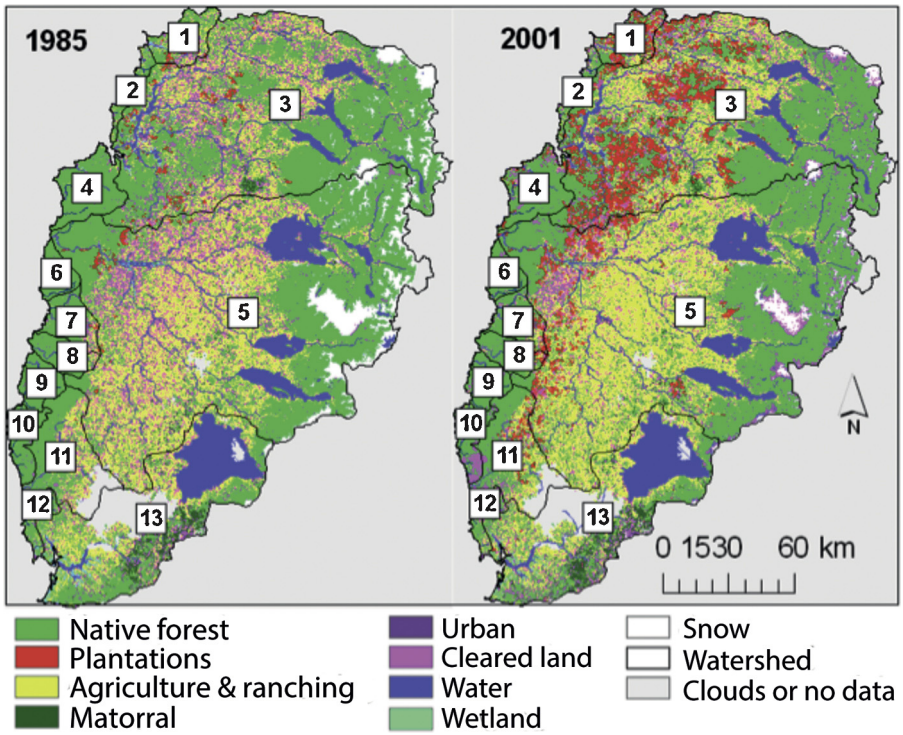

Fig. 1. Plantations were established predominantly on the foothills of agricultural watersheds, rather than watersheds dominated by native forest or flat agricultural lands. The supervised classification of October 5, 1985 (Landsat TM, path 233, rows 87–89) and November 29, 2001 (Landsat ETM, path 233, rows 87–89) images shows native forest, plantation, matorral (shrubland), agriculture, cleared land, sand, water, urban, snow, and wetland land cover/uses. The numbers in white boxes correspond to watersheds in Fig. 2. Overall accuracy in 2001 is 90% with a 0.90 kappa coefficient; average 1985 classification accuracy is 91% with a 0.90 kappa coefficient (see Appendix A for full accuracy assessment). The plantation watersheds are numbers 1, 3, 5, 7, and 11.

where they are established and likely do not represent the same drivers as reforestation of native forests.

1.1. Can the forest transition framework explain plantation establishment?

Forest-transitions research focuses on explaining natural, native forest regeneration (Rudel et al., 2005), although planted trees are included in some forest-transition studies (Mather, 1992; Mather, 2007; Farley, 2007; Grau et al., 2008; Meyfroidt and Lambin, 2008; Rudel et al., 2009; Lambin and Meyfroidt, 2010; Aide et al., 2012). This study contributes an analysis of an important case study to the growing research on tree plantations and forest transitions. The Forest Transitions Framework (FTF) explains many observed temporal and spatial patterns of native forest regrowth despite continued deforestation at the global level. According to the FTF, less-productive land becomes reforested as rural populations abandon their fields and migrate to urban areas where economic opportunities have grown (Mather, 1992; Rudel et al., 2005; Lambin and Meyfroidt, 2010). The FTF suggests that agriculture plays a role in tree-plantation establishment because tree-cover increases commonly occur on land previously cleared for agriculture as agricultural regions are concentrated and intensified on optimal lands (Mather, 1992; Rudel et al., 2005; Lambin and Meyfroidt, 2010). If the FTF explains tree plantation patterns, then we would expect tree plantations to develop in areas where rural farmers have moved to urban areas and abandoned their fields that regrow. We would also expect that tree plantation crops replace other agricultural crops or marginal agricultural lands, rather than native forests. Within the forest-transitions literature, Chile is often cited as a case in which tree plantations are established (Rudel, 2009). Chile was the world's sixth largest wood pulp exporting country by 1992 (Sedjo, 1999). In 2005, plantations comprised 2.1 million hectares or 13.4% of the total tree cover (INFOR, 2006). Using satellite image analysis, Echeverria et al. (2006) and Aguayo et al. (2009) concluded that mainly substitution – of native forests for plantation is occurring in central Chile. In contrast, Clapp concludes from interviews with small landholders that plantations in central Chile were a result of planting trees on lands that were previously agricultural areas (Clapp, 1995, 2001). Using geomorphology, soil characteristics, and road networks, Wilson et al. (2005) identified sites in southern Chile that would be vulnerable to plantation establishment; they assumed that most tree plantations were a result of direct substitution of native forest for tree crops.

1.2. Will aggregating tree-plantation and native-forest classes mask the conclusions drawn from land cover change analyses?

Tree plantations and native forests are different ecologically, so how does treating them as one class confound analyses and conclusions? In comparison to native forest, tree plantations are associated with lower biodiversity (Munoz et al., 1987; Vergara and Simonetti, 2004; Saavedra and Simonetti, 2005; Carnus et al., 2006) and biodiversity potential (Hall et al., 2012). Carbon sequestered in tree plantations established through substitution usually decreases (Fearnside, 1995; Glenday, 2006; Nosetto et al., 2006; Betts et al., 2007) and plantations were associated with lower carbon sequestration potential (Hall et al., 2012). Watershed hydrology changes when plantations are established because typically tree plantations have higher evapotranspiration rates than native forests or grasslands/cultivated lands, resulting in less water flow to catchment streams (Farley et al., 2005; Huber et al., 2008). Erosion and consequent sediment loading in streams are high at the onset of plantation establishment and after harvesting (Stevens et al., 1994; Oyarzun and Pena, 1995; Farley and Kelly, 2004; Oyarzun et al., 2007) and trees planted on higher slopes lead to higher erosion

and runoff potential (Schlatter, 1977). Higher nitrogen levels occur in streams downstream from tree plantations (Oyarzun et al., 2007) and other changes in water quality occur (Farley et al., 2008).

Although native forests and tree plantations supply distinctly different ecosystem services, many studies typically use the aggregated "tree cover" as a variable to model land-cover effects on hydrological processes (Murray et al., 2013), carbon dynamics (Dong et al., 2003), and stream integrity (Stevens et al., 1994). Forest-transitions research has been developed with FAO datasets that aggregate tree plantations with native forest classes. Only in 2005 did the Forest Resources Assessment begin to clarify definitions of tree plantations vs. native forests among participant countries; however, these data are still at the country-level resolution (Mather, 2007); and today aggregated classes are still used (Viña et al., 2016). Hansen et al. (2013) recently produced global high resolution (900 m^2), spatially explicit datasets and maps of disturbance. These maps identify areas that have regrown after a disturbance, which may indicate plantations, but these datasets may not be sufficient to understand patterns of tree-plantation establishment because they may aggregate plantations and natural forest regrowth.

We tested the following hypotheses derived from the FTF with a combination of land-use/land-cover change maps derived from satellite images, national soil-use maps, national land-cover maps, slope data, digital elevation models, and census data.

H1. Plantations will become established predominantly in watersheds dominated by agriculture rather than watersheds dominated by native forest.

H2. Plantations will become established at the foothills of the mountains, with moderately steep slopes and mid-level elevations (marginal agricultural lands) as opposed to flat, agriculturally intensive regions in the valleys (low elevation).

H3. Most plantations will be established on agricultural areas, rather than replace or substitute native forest.

H4. Substitution of plantations for native forest will diminish over time.

H5. Areas with more plantations will be associated with rural population loss.

Next we ask whether aggregating tree plantation and native forest tree covers distorts the result that tree plantations negatively influence the quality of nearshore resources. Fishers harvesting nearshore marine resources downstream of catchments with tree plantations caught lower-quality shellfish and were paid lower prices for their catch (Van Holt, 2012). They show that phytoplankton biomass in the receiving nearshore ecosystem is higher, most likely due to nitrogen inputs, which in turn increased the amount of parasitic organisms growing on the commercially valuable mollusk *loco* (*Concholepas concholepas*) shells, thus reducing their size (Van Holt et al., 2012). We test whether aggregating native forest and tree plantation classes (i.e., examining total tree cover) in models distorts these conclusions.

2. Materials and methods

2.1. Study site

We focus on Chile because it is a leader in global pulp production from tree plantations. We selected thirteen watersheds at the southern frontier of plantation development, along 250 km of coast (spanning 35,853 km^2, about the size of The Netherlands). The study site contains three large watersheds [Valdivia (3), Osorno (5), and Llanquihue (13)] and ten small watersheds (1,2,4,6,7,8,9,10,11,12) (Fig. 1). These comprise 27 counties and two administrative regions (region XIV, Los Rios, and the northern portion of region X, Los Lagos; Chile numbers its administrative regions with Roman numerals).

2.2. Image preprocessing and analysis

Three October 5, 1985 Landsat 5 TM scenes and three November 29, 2001 Landsat 7 ETM+ scenes were selected (WRS II path 233, row 87–89), to correspond to the spring season. The 1985 images were acquired 11 years after the 1974 forestry law that began subsidizing plantations with government grants and tax breaks (Lara et al., 1993) and the 2001 images represent the landscape several years after Chile doubled its pulp export capacity, which established Chile as a global pulp producer (Sedjo, 1999). A 1995 land-cover map (*catastro*) generated from aerial photographs (CONAF et al., 1997) was used as an additional source of training samples. The images were georectified (rms error < 15 m) and atmospherically corrected using the Center for the Study of Institutions, Populations, and Environmental Change (CIPEC) protocol that includes converting radiance values to surface reflectance values and dark object subtraction (Markham and Barker, 1986; Teillet and Fedosejeus, 1995). The low-gain portion of band 6, the thermal band, was converted to at-sensor temperature or black body values since these values have been shown to be highly correlated to land-cover type (Southworth, 2004). All scenes were topographically corrected using a modified Minnaert Correction according to Teillet et al. (Minnaert, 1941; Teillet et al., 1982), using the Space Shuttle Radar Topography Mission (SRTM) Finished 3 arc-second (90 m) raster elevation dataset and the derived slope image. The atmospherically and topographically corrected images were transformed using the tasseled-cap transformation, which developed six new data axes that represent brightness in the image land covers, greenness in the vegetation, wetness which corresponds to vegetation health, and fourth, fifth, and sixth layers (Kauth and Thomas, 1976). A normalized differential vegetation index layer [NDVI:(reflectance band 4–reflectance band 3)/(reflectance band 4 + reflectance band 3)] (Rouse et al., 1974) was created. Finally a texture analysis was run on the mid-infrared Landsat band 5 to create three new layers: a range, mean, and standard deviation texture layer. Band 5 was used because it was most useful in separating tree plantations from native forest in trial tests.

2.3. Land cover/use change analysis

From July to August in 2004 we field-surveyed 169 sites for their land cover/use for accuracy assessments. At each site, we listed the main tree species present, percent ground cover (measured by taking a visual inspection of approximately a 10 × 10 m plot and estimating the amount of ground cover), and canopy closure (measured by visually inspecting approximately a 30 × 30 m area and estimating the percentage of light that will go through the entire area). We also estimated the age of the tree plantations or forest measured by counting the rings of the stumps, using a reference of growth of approximately 4 cm each year at breast height for pine and 2 cm each year for eucalyptus (Toral et al., 2005; Munoz et al., 1987) and knowledge of land use in the region.

We used a maximum likelihood supervised classification method to create nine land cover classes (1) native forest, that is, both primary and secondary forests of Siempreverde (includes olivillo costero and alerce forests) and Roble-Raulí-Coihue (includes *Nothofagus* deciduous and *Nothofagus* evergreen forests) forest types (2) plantation that included exotic pine (*Pinus* spp.) and eucalyptus (*Eucalyptus* spp.), and rarely observed Douglas Fir[1] (*Pseudotsuga menziesii*) (3) shrubland or *matorral*, which here we characterize as a mosaic of the natives *Chusquea* spp, *maqui* (*Aristotelia chilensis*), *Baccharis* spp, *Blechnum* spp, and the exotics *ulex* (*Ulex europaeus*) and *retamilla* (*Teline monspessulana*) (4) agri-

[1] We classified tree plantation cover; any tree plantations that were cleared in the satellite image were classified as cleared land

culture, including cattle ranching, (5) cleared land[2] (6) urban, (7) wetland, (8) water, and (9) snow classes. There was some spectral confusion between the native forest and plantation pixels, which was typically in the middle of elevated areas (Andes mountains or coastal forest). We therefore divided the study area into two and ran separate classifications using the same spectral signatures on each section. One slice of the image contained high elevation areas that included the Andes and coastal forest mountains and likely contained mainly native forest. The second slice was more likely to be plantation areas. Here we broadly selected regions that were already plantation polygons from the national vegetation map (CONAF et al., 1997) and included other regions that contained potential tree plantations (most tree plantations were distinguishable in the satellite image). We then used the same spectral signatures to classify these scenes independently, and joined them together. Both areas contained native forest and tree planation land cover types and this solved the spectral confusion in the mountainous regions.

We further refined the classification through post-processing. We reclassified pixels that were clearly plantations, i.e., those pixels located in the middle of plantations that were classified as cleared land in 1985 and native forest in 2001. These pixels were likely plantations as native forest types are slow to regrow and are not located inside plantations. Second growth forest may only initially recruit after five years of a clearcut (Dezzotti et al., 2004). *Drimys winteri* and *Nothofagus pumilio* take ten years to reach 2, 3, and 4 m, respectively (Drake et al., 2003). *N. obliqua*, *N. alpina*, and *N. dombeyi* reach 4 m after ten years (Donoso et al., 1993). *Fitzroya cupressoides* grows even slower. There was no evidence that we classified these regrowth areas incorrectly based on field visits to five sites and the accuracy assessment of the image. Also, we reclassified the minimum size of the urban class to be ten pixels, which reduced confusion between cleared land and urban areas.

The field survey data (169 site visits) were used to generate accuracy samples. A second accuracy assessment was conducted by generating 30 random points in each land cover class to capture the accuracy of a location that may not be easily accessible by road. In this case we assessed accuracy using the additional datasets that include 1) the national vegetation map (CONAF et al., 1997) that contained detailed forest and tree-plantation age information and agricultural information and 2) satellite image interpretation i.e. we could easily distinguish cleared land and agricultural areas on a Landsat band 5-4-3 (RGB) composite image and plantations were visually distinct from native forest when inspecting a 5-4-3 composite of the satellite image. We calculated the kappa coefficient, the user accuracy, and producer accuracy for each accuracy assessment.

For H 1–3, after the supervised classification, area (km^2) and percentage of each land cover/use class (km^2 of LCLU/total watershed area $km^2 \times 100$) in 1985 and 2001 was calculated in each of the thirteen watersheds. Then, the total area (km^2) and percentage of each class that converted from the 1985 class to the 2001 class was calculated for each watershed. Here we focus on watersheds because environmental management is often at the watershed region and land-aquatic interactions are relevant at the watershed scale. For H4, we evaluated the entire study site as a whole as we were looking for very broad patterns. For H5, we were interested in social processes and ran county-level data to match census surveys.

2.4. Landscape change patterns

To test **H1**, first land-use trajectories were created to show the change in the percentage of tree plantations, native forest, and agriculture/ranching in the 13 watersheds from 1985 to 2001. Watersheds then were grouped into three classes based on the percentage of land area that was converted to plantations (from 1985 to 2001), the percentage of native forest in the watershed (both in 1985 and in 2001), and the percentage of agriculture/ranching in the watershed (both in 1985 and in 2001). Second, a principal components analysis (PCA) was run on the percent of native forest, plantations, agriculture/ranching, and cleared land in the 13 watersheds in 1985 and 2001. Bi-plots (Gabriel, 1971; Gower and Hand, 1996) of the first two principal components (PC1 & PC2) identified the main land cover components, which were subsequently used to understand change patterns. To test **H2**, agricultural, tree plantation, and native forest areas in 2001 were summarized by elevation and slope layers and by soil-use type from the soil-use map (CIREN and CORFO, 1983). Soil classes were defined by the Centro de Información de Recursos Naturales (CIREN) in terms of their capacity for use and erosion potential. Higher numbers indicate steeper land, often at higher elevations, and land with more erosion potential and decreasing agriculture potential (CIREN and CORFO, 1983). Class II through III soils are classified as appropriate for agriculture with few limitations on use. Schlatter, (1977) argues that pine plantations are best suited for moderately sloped, poorly drained soils (classes IV–VI) and highly sloped soils (class VII), though he argues that steeply sloped soils be preserved for natural forest due to the high risk of erosion. Eucalyptus trees grow on similar soils but grow better on type IV–VI soils because they cannot withstand cold temperatures at higher elevations. To test **H3** the area of tree plantations in 2001 that were bare soil or agriculture in 1985 was calculated. That area was compared to the area of tree plantation in 2001 that were classified as native forest in 1985.

To test **H4** we included an additional dataset, the 1995/96 land-cover map of Chile to calculate the area and percent of tree plantations that were converted from native forest from 1985 to 1995 and from 1995 to 2001.

To test **H5**, the change in rural population from 1994 and 2002 per county was calculated using the Chilean Census Data (INE, 1994, 2002) and these values were compared to the percent and area of tree plantations that were established per county from 1985 to 2001. An ANOVA was used to compare the mean percent population loss per county for plantation counties (>10 km^2 plantations established from 1985 to 2001) and non-plantation counties ($\leq$10 km^2 plantations established between 1985 and 2001). The threshold of 10 km^2 was chosen so that counties with a plantation region were compared to those that had essentially no plantations (see Fig. 2.8 in Van Holt (2012). This analysis was run two ways, first with all counties and second, excluding counties with growing rural populations, because these counties are usually urban centers that have spillover populations to rural areas. Then a regression model was created with social and landscape drivers at the county-level unit of analysis to determine which factors best-predicted plantation establishment. Dependent variables in the regression were percent conversion of native forest and agricultural areas to plantations from 1985 to 2001. Independent variables included percent rural population loss from 1994 to 2002, total area of land that remained as native forest from 1985 to 2001, total area of land that remained in agriculture from 1985 to 2001, and the total potential usable area, which was derived from soil use maps (water and rocky locations were not suitable for plantations and were subtracted from a county's total area). This analysis was limited to counties with at least 10 km^2 of plantation establishment from 1985 to 2001.

To show how aggregating tree plantations and native forest affects model outcomes, first we ran six individual regressions with percent phoronid or barnacle cover on loco shellfish as the dependent variable and with percent native forest cover in 2001, percent tree plantation cover in 2001, and total tree cover in 2001, which is

[2] We did not distinguish whether cleared land was due to traditional agricultural practices, deforestation, or tree plantation establishment.

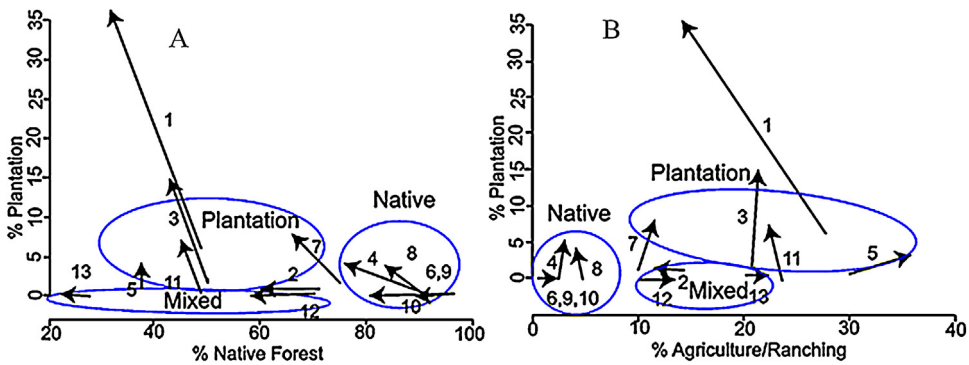

Fig. 2. Trajectory plots of the change in each watershed from 1985 to 2001 show the relative percent of (A) plantation and native forest and (B) plantation and agriculture/ranching classes. Each numbered arrow represents a watershed from Fig. 1 and begins at the percentage of each pair of land covers in 1985 and ends at the paired percentage of land covers in 2001. The ellipses identify watersheds with similar land cover/use change trajectories.

an aggregated percent native forest and tree plantations metric, as independent variables. We also compared the results of canonical correlations of land and seascape variables and loco shellfish characteristics. The first correlation included used only tree plantations in the watersheds nearby fisheries management areas whereas the second correlation used aggregated total tree cover variables. All other variables were the same in both analyses and included landscape and seascape variables: percent change in agriculture, cleared land, urban land class, matorral change, and average chlorophyll a characteristics in the fisheries management area. These variables were correlated with characteristics of 30 loco shellfish that were sampled from June and August of 2003 in each management area (N = 42) and included: percent cover on locos of phoronids, shell-boring bivalves (*Hiatella solida*), polychaetes (*Dodecaceria* cf. *opulens*), barnacles, shell height, loco meat weight, shell length, and percent lipid in loco meat (Van Holt et al., 2012).

3. Results and discussion

3.1. Plantations replaced subsistence agriculture in agriculturally-dominant areas

Plantations became established in agriculturally-dominated watersheds, rather than in watersheds dominated by native forests, and in marginal foothills rather than on flat agricultural areas, supporting **H1** and **H2**. Plantations are a growing proportion of the landscape, at the cost of native forest in several watersheds. They covered 2266 km^2 (6%) of the study region in 2001, ranging from 0 to 29% depending on the watershed, a 1991.8 km^2 increase from the 1985 area (Fig. 1; Appendix B). The 1985 classified image showed that out of 16,593 km^2 of tree cover area, 16,319 km^2 was natural forest and only 274 km^2 was plantation. Native forest decreased by 1298 km^2 between 1985 and 2001, during which agriculture increased by 855 km^2. Cleared land decreased (−695 km^2) from 1985 to 2001, about half of which converted to plantations in 2001 (Appendix B).

H1 was supported because watersheds with more than 80% native forest cover in 1985 that did not transition to plantation watersheds. Trajectory plots (Fig. 2) of class changes in each watershed from 1985 to 2001 of the percent of tree plantations and native forest (A) and percent tree plantation and agriculture/ranching (B) identify three distinct types of watersheds—*native forest*, *plantation*, and *mixed*. *Plantation watersheds* (1, 3, 5, 7, and 11) (Figs. 1 and 2) had moderate to large increases in plantation fraction (3% to 35%) between 1985 and 2001 (watershed 5 is large so the fraction is relatively small). In 2001, these watersheds had typically low native forest fraction (below 50% except for watershed 7 which was 67%) and high agriculture fraction (10–35%) with moderate loss in fraction of native forest (3–13% loss, except watershed 5, which remained relatively unchanged) from 1985 to 2001. In contrast, *native-forest watersheds* (4, 6, 8, 9 and 10) started with a very high (>80%) fraction of native forest, which they maintained (>78%) in 2001. They had a low (<5%) fraction of cleared land in 1985 and shifted to low to moderate (6%-15%) fractions of cleared land in 2001. Agriculture fractions were low and ranged from 1% to 4% in 2001. In the *mixed watersheds*, plantations were minimal in both years.

The PCA also supports **H1** because plantations positively covary with agriculture and cleared land and were negatively related to native forest cover (Fig. 3). This means that watersheds with high native forest cover had smaller agriculture, plantations, and cleared land areas. The watersheds group in similar patterns in 1985 and 2001 regardless of plantations, demonstrating that percent native forest and agriculture have a larger role in explaining land-use patterns. PC1, which is loaded mainly on the proportion of native forest in each watershed, explains the largest fraction of variance in land cover/use characteristics per watershed in 1985 and 2001 (79% in 1985 and 70% in 2001; eigenvalue ≥0.90). PC2 explains an additional 17% (1985) and 16% (2001) of variation in the LCLU in the watersheds. The main distinction between PCs in 1985 and 2001 is that plantations became important factors of PC2 in 2001 (Appendix C, 2001 PC2 eigenvalue −0.92 compared to −0.21 in 1985).

H2 is supported because areas that were converted from agriculture to tree plantations had an average slope of 8°±5° (s.d.) and elevation of 228 m ±114 m whereas areas in sustained agricultural land covers were flatter and at lower elevations (average slope of 3° ± 4° and elevation of 152 m ± 98 m). Subsistence agriculture occurred on higher slopes. Only a small fraction of land where tree plantations were established (4% in 1985 and 5% in 2001) was classified as suitable for intensive agriculture (Chile Agricultural Ministry Classes II and III) (Appendix D). Class VII type soils were mainly used for tree plantations accounting for 63% (1985) and 69% (2001) of tree plantation cover. About half of the land that remained in agriculture was classified in Class I–III soil types and half was classified in soils IV, V, VI, VII, and VIII. Since agriculture expanded less than plantations (accounting for only approximately 30% increase rather than the approximate 70% that is accounted for by plantation conversion) and Chile's agricultural output continues to grow, agriculture and dairy activities throughout the study site have probably become intensified at the same location since 1985. Native forests converted to plantations were on lower slopes (10° ± 5°) and elevations (278 m ± 132 m) whereas the remaining native forests were at higher slopes (13° ± 9°) and elevations (562 m ± 353 m). These native forests likely had several rounds of selective logging, ending in firewood extraction. All of this is combined with browsing by cattle.

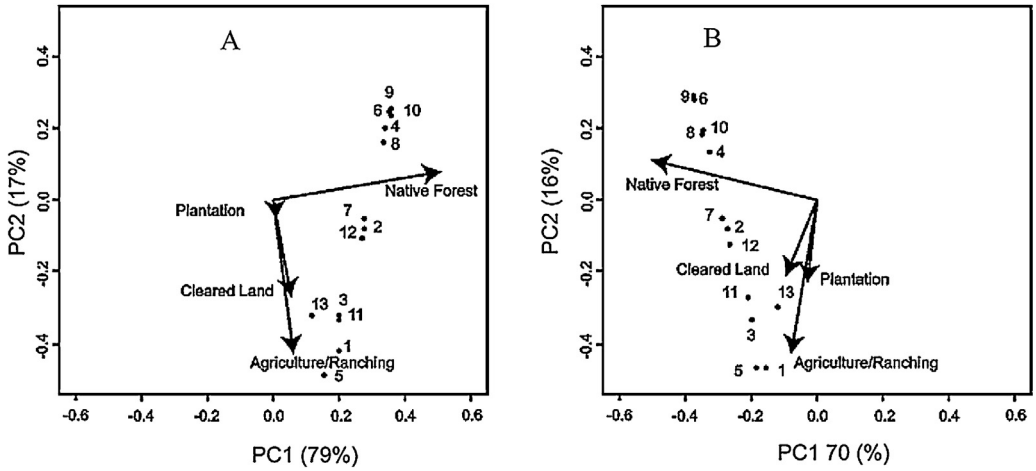

Fig. 3. First principal component (PC1) and second (PC2) principal component for (A) 1985 and (B) 2001 land cover fraction variables. The labels correspond to watersheds in Fig. 1. The percentages are variance accounted for in the corresponding component.

These findings agree with the results of forest transition studies (Mather, 1992; Rudel et al., 2005) and other plantation land-cover classification studies in Latin America (Lira et al., 2012; Redo et al., 2012) because plantations have been associated with agricultural regions although they do not replace intensive agriculture. Plantations were not established in the site with intensive agriculture in the Atlantic Forests of Brazil (10% native forest cover and ~80% agriculture); instead plantations were established in sites with 30% and 50% native forest cover and ~40% agriculture, which are similar percentages of the landscape that was covered in native forest and agriculture where plantations were established in our study (Lira et al., 2012). Plantations are decreasing in Uruguay because industrial-scale agriculture has quadrupled from 2000 to 2010 and subsidies have decreased for plantations (Redo et al., 2012). Agriculture is likely a more profitable land use for this region and since agriculture is increasing, there is no opportunity for plantation establishment. This finding reinforces our conclusion, and the prediction from the FTF, that plantations were not established in areas of intensive agricultural development.

3.2. Deforestation was associated with tree plantations

Native forests were converted, or more accurately, deforested and then replanted with plantation trees, to establish plantations. **H3** and **H4** were rejected. Tree plantations are business enterprises tied to global markets, whereas reforestation of native forests or planting trees for local wood demand as predicted by the FTF occurs when people abandon their lands. Plantations were mainly established on land that was previously agricultural or bare soil areas accounting for 1067 km^2 or 47% of all plantation transformations, substitution of native forest closely followed (accounting for 44% or 983 km^2) (Appendix B). The other 9% (211 km^2) were already established as plantations in 1985. Of course, even agricultural areas that now are tree plantations were once native forest.

Native forests were still being converted to plantations from 1995 to 2001 (not supporting **H4**). Out of the 983 km^2 of native forest that people converted to plantation between 1985 and 2001, 562 km^2 (57%) of the native forest was already converted to a plantation in 1995 and the remaining 308 km^2 of native forest (31%) was converted since 1995. Controlling for year, the amount of tree plantations planted on average per year from 1985 to 1995 was 56.2 km^2 and from 1995 to 2001 was 51.3 km^2, which is essentially equal. Other researchers have also found that deforestation is associated with tree-plantation establishment (Carlson et al., 2012; Hansen et al., 2013).

Table 1
Regression analysis of percent of the county with native forest and agriculture convert to plantations from 1986 to 2001 (PLANT EST) (dependent variable) and percent rural population loss per county (POP LOSS) (YEARS), total area that remained in native forest land cover from 1986 to 2001 (STBL NAT), total area that remained in agriculture from 1986 to 2001 (SBL AG), and total usable area (USE AREA) (total area-water and rocky landscape as defined by soil-use maps) (CIREN and CORFO, 1983). $R^2 = 0.663$, $p = 0.007$.

Independent Variable	UnStandardized Coefficients	Beta	p-value
(Constant)	15.6	>0.00	
POP_LOSS	0.1	0.42	
STBL NAT	−0.03	0.02	
STBL AG	−0.03	>0.00	
USE_AREA	0.02	0.04	

3.3. Population loss does not predict plantation establishment

The regression analysis showed that **H5** was not supported. The percent of native forest and agricultural land cover/use that converted to plantations from 1985 to 2001 did not depend on the loss of rural populations ($R^2 = 0.663$, $p = 0.007$) (Table 1). The area of native forest and agriculture that remained the same from 1985 to 2001 was negatively associated with plantation establishment, reconfirming the finding that tree plantations are not established in watersheds dominated by native forest and that they do not replace intensive agriculture. Tree plantation establishment was related to the total amount of useable area, likely the marginal regions. There was no difference in the mean rural population loss in plantation and non-plantation counties (two treatments included all counties and excluded counties with any population increase). This conclusion was also found by Izquierdo et al. (2008); tree plantations in Argentina were not necessarily associated with rural population loss because the Argentine Candelaria and Eldorado counties had an increase in rural population and tree plantations

3.4. Aggregating tree plantations and native forest into total tree cover in analyses masks effects

The percent of tree plantations in 2001 (%PL01) in the watershed that flows into the fisheries management area was positively associated with a higher percentage of barnacle (*Verruca laevigata* and *Jehlius cirratus*) and phoronid (*Phoronis ovalis*) organisms that bore into the loco shellfish (% phoronids = $-0.92 + 1.21 \times$ %PL01 ($R^2 = 0.67$, p-value < 0.0001); % barnacles = $25.8 + 0.805 \times$ %PL01 ($R^2 = 0.14$, p-value = 0.0135)), which have been shown to cause fishers to receive a lower price for their catch (Van Holt, 2012). There

was no statistically significant relationship between the percent of phoronids or barnacles on loco shells when regressed with an aggregated percent total tree cover variable that included both native forest and tree plantations in 2001.

The average chlorophyll-*a* characteristics in the fisheries management area (CHLA) had a high correlation coefficient in the canonical correlations between land and seascape variables and shellfish quality (see Appendix E for complete canonical correlation). When tree plantations were isolated in the analysis, the increase in plantations (PPL_CH) in nearby watersheds was highly correlated with lower shellfish quality correlates (percent phoronids (FOR) and barnacles (PICTOTAL) on loco shells). However, when aggregated total tree cover area (TCOV_CH) and percent (PTCOV_CH) tree cover change variables were used, the correlation showed that an increase in tree cover area was related to shellfish-quality decline. The total tree cover variables are likely picking up on the influence of tree plantations, but aggregating to a total tree cover variable masks the real variable of interest, tree plantations.

3.5. Plantation transitions

We suggest the following modifications to the FTF to develop a series of testable hypotheses that focus on plantation transitions. In line with the FTF, (i) plantation transitions are more likely to occur on the foothills or on subsistence lands where the agricultural lands are not as productive. (ii) Agriculture is an important component that predicts establishment; it is more likely that plantations will be established in agriculturally-dominant areas rather than in areas dominated by native forest.

Plantation transitions also have characteristics that are not in line with the FTF. (iii) Native forests are also deforested to establish tree plantations. Investigators should indicate whether plantations are a result of replacing agricultural areas, native grasslands, or substituting native forest and over what timeframe. (iv) If tree plantations are included in forest transitions, then there is a mismatch in how well demography predicts transitions. Plantation transitions are not always associated with rural population loss. These areas are not in natural forests that are growing when people abandon fields; this is in line with research in Ecuador, where plantations were actively established for economic purposes, to develop the land, rather than to abandon land (Farley, 2007). (v) Tree plantations are commodities and their expansion is connected to global markets. We expand on Rudel et al. (2005) pathways of forest transitions that state that transitions will occur when people abandon agriculture and native forest regenerates and when low native forest cover and market demand drives people to plant trees, including plantations. Indeed, Meyfroidt and Lambin, (2008) indicated that two forest transition pathways do not precisely explain all plantation establishment, though their alternative pathway focused on small landholders, rather than companies. Commercial operations in Chile have little to do with supplying forest products for local markets or to protect watersheds from erosion or rebuilding tree cover. Plantation development exploded with the "Decreto Ley 701" law of 1974 when the Pinochet administration subsidized 75% of the cost of plantation establishment and eucalyptus (*Eucalyptus globulus* and *Eucalyptus nitens*) became an important pulpwood species (Lara et al., 1993). In volume, in the study area, around 70% of the forest production went to pulp and paper, 25% to sawn timber, and 5% to other products (INFOR, 2006). The government subsidized in-demand non-traditional agricultural exports such as pine trees to bring Chile into the world market economy and to make the country less vulnerable to the fluctuations in the copper market (Auty, 1993; Gwynne, 1996). (Mather, 2007)Mather's modification to the FTF emphasizes that transitions in the global south are different because incomes at the national level are low (transitions initially were discussed in Europe). Likely this rise in tree cover is, in part, due to tree plantations, that are established in the global south. (v) Plantation companies seek areas to develop plantations where the financial rewards are likely to be the highest. Tree plantations represent an expansion of, not a reduction of agriculture.

3.6. Future research

We conducted a hypothesis-driven analysis to test how well FTF explains frontier tree planation establishment in Chile. Land-cover change processes in the country as a whole or across other countries may require future work. For example, in other countries or region, if a strong agricultural and forestry sector is not present, plantations may be associated with different landscape characteristics. Likewise if tree plantations already dominate in a region, the same patterns may not appear. Native forests are sometimes converted to tree plantations because native forests are deemed degraded; however, the definition of a degraded forest, and how this varies across countries and municipalities and in practice is unclear and this is a topic for future work. Understanding which forest types are more prone to deforestation could be an important first step. The types of agriculture and labor associated compared to the labor force of tree plantations could offer new insights into the migration patterns that we find here and in other studies. Likewise understanding the role of land tenure could be important to consider in the future. To understand how the FTF can be applied across the globe, we need a systematic meta-analysis of tree-plantation studies to test these predictions further.

4. Conclusion

Forest transitions explain some patterns of tree plantation development—plantations replace subsistence agriculture, intensive agriculture remains relatively intact, and plantations become established in agriculturally-dominant regions. In contradiction to the FTF, tree plantations often destroy native forest for their establishment. Rural outmigration does not explain plantation establishment. Tree plantations are more like agricultural crops because they are harvested periodically. They are associated with native-forest loss. They represent an expansion, rather than a reduction in agricultural area. Rural people are not leaving prior to tree plantations and in some areas rural populations are increasing. In the tree plantation frontier in Chile, the foothills, which are subsistence agriculture areas, in agriculturally-based watersheds, are most threatened for conversion to tree plantations; similar marginal landscapes may be likely regions for tree plantations in the global south. To make sure that we fully understand the role of plantations in the ecosystem and society, we should treat them as a land cover/use independent of native forests.

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Appendix A. Accuracy assessment for the supervised classification of the November 29, 2001 (Landsat ETM, path 233, rows 87–89) and the October 5, 1985 (Landsat TM, path 233, rows 87–89) satellite images. Accuracy was assessed with 1) targeted field visits to 169 locations and 2) 30 randomly selected points for each class using secondary data sources. User accuracy of 94% means that of the pixels classified as plantations 94% are indeed plantations, while the producer accuracy means that 87% of the pixels were classified as plantations

Class Name	Targeted field-based accuracy assessment				Randomly generated desktop accuracy assessment			
	REFERENCES Total Pixels	Classified Total Pixels	Producer Accuracy%	User Accuracy%	Reference Total Pixels	Classified Total Pixels	Producer Accuracy%	User Accuracy%
2001 Image								
Plantations	362	337	87	94	30	31	93	90
Native Forest	286	292	85	83	30	27	80	89
Shrubland	266	217	78	95	30	35	97	83
Agriculture	159	188	100	85	30	27	87	96
Cleared land	210	226	91	85	30	23	70	91
Water	50	50	100	100	30	32	100	94
Urban	292	272	93	100	30	32	100	94
Snow	46	46	100	100	30	30	100	100
Wetland	65	50	77	100	30	33	100	91
1985 Image								
Plantations	384	311	81	100	30	28	93	100
Native Forest	256	318	100	81	30	26	80	92
Shrubland	238	191	80	99	30	36	93	78
Agriculture	234	293	100	80	30	29	87	90
Cleared land	230	266	99	86	30	28	70	75
Water	246	250	98	97	30	30	100	100
Urban	97	76	78	100	30	30	97	97
Snow	261	256	97	99	30	30	100	100
Wetland	223	184	81	98	30	33	90	82

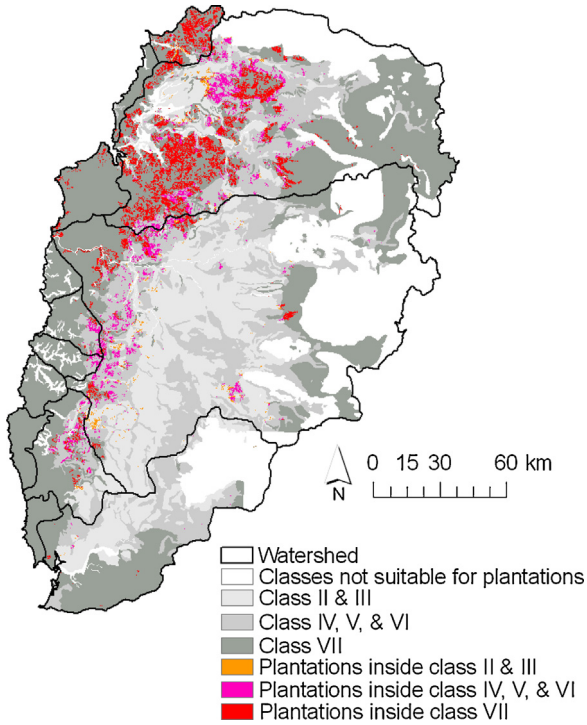
Appendix B. Area (km^2) and percent of change and 2001 land use/cover of plantation, native forest, *matorral* (shrubland), agriculture, and cleared land class for each watershed as labelled in Fig. 1. Area change was determined by calculating the difference in area between respective land cover class in the November 29, 2001 image (Landsat ETM, path 233, rows 87–89) and the October 5, 1985 image (Landsat TM, path 233, rows 87–89) for each class and each watershed. Area in any land-cover type for 1985 can be calculated by subtracting the change from the value for 2001. Percent change between 2001 and 1985 was calculated by dividing the area change by the total watershed area

Watershed	Plantation		Native Forest		Shrubland		Agriculture		Cleared Land	
#	Change	2001	Change	2001	Change	2001	Change	2001	Change	2001
Area (km^2)										
1	151.0	181.9	−89.9	169.2	0.1	2.0	−69.2	76.6	10.0	90.0
2	0.4	2.5	−20.4	140.8	0.0	0.1	−4.6	27.4	25.4	43.6
3	1264.0	1425.0	−677.3	4483.5	−7.5	73.4	61.0	2172.5	−345.7	1365.5
4	27.1	27.1	−109.6	591.0	0.5	0.5	2.0	21.0	78.3	101.3
5	417.2	489.5	56.8	5815.1	−8.7	155.0	836.1	5403.4	−688.6	1855.3
6	0.0	0.0	−14.6	268.1	0.1	0.1	0.9	4.4	13.1	16.7
7	27.0	33.1	−37.4	325.7	0.6	2.1	4.6	54.0	4.0	66.2
8	12.2	12.2	−24.4	313.7	0.9	1.0	−1.6	16.0	11.8	24.4
9	0.0	0.0	−18.6	348.4	0.1	0.1	0.8	3.8	17.1	21.9
10	0.0	0.0	−96.6	629.6	0.1	0.2	1.4	10.2	93.5	110.0
11	84.9	86.3	−45.0	637.1	1.5	4.9	−14.1	313.1	−27.1	152.8
12	0.0	0.0	−28.1	158.1	3.1	4.0	6.9	33.8	18.4	64.6
13	8.0	8.0	−192.4	1141.2	68.5	308.7	30.6	990.9	94.9	634.9
Total	1991.8	2265.6	−1297.5	15021.5	59.3	552.1	854.8	9127.1	−694.9	4547.2
Percent										
1	28.7	34.5	−17.1	32.1	0.0	0.4	−13.1	14.5	1.9	17.1
2	0.2	1.1	−9.0	62.4	0.0	0.0	−2.0	12.1	11.2	19.3
3	12.3	13.9	−6.6	43.7	−0.1	0.7	0.6	21.2	−3.4	13.3
4	3.6	3.6	−14.4	77.7	0.1	0.1	0.3	2.8	10.3	13.3
5	2.7	3.2	0.4	37.9	−0.1	1.0	5.4	35.2	−4.5	12.1
6	0.0	0.0	5.0	91.2	0.0	0.0	0.3	1.5	4.4	5.7
7	5.6	6.8	−7.7	67.1	0.1	0.4	1.0	11.1	0.8	13.6
8	3.3	3.3	−6.6	84.6	0.2	0.3	−0.4	4.3	3.2	6.6
9	0.0	0.0	−4.9	91.9	0.0	0.0	0.2	1.0	4.5	5.8
10	0.0	0.0	−12.7	83.0	0.0	0.0	0.2	1.3	12.3	14.5
11	6.1	6.2	−3.2	45.7	0.1	0.3	−1.0	22.5	−1.9	11.0
12	0.0	0.0	−10.6	59.6	1.2	1.5	2.6	12.7	6.9	24.3
13	0.2	8.0	−4.0	23.8	1.4	6.4	0.6	20.7	2.0	13.3
Total	5.6	6.3	−3.6	41.9	0.2	1.5	2.4	25.5	−1.9	12.7

Appendix C. Eigenvalues and eigenvectors of PCA1, PCA2, PCA3, and PCA4 from the principal components analysis (PCA) on the percent of native forest, plantations, agriculture/ranching, and cleared land classes in the 13 watersheds derived from a supervised classification of the November 29, 2001 image (Landsat ETM, path 233, rows 87–89) and the October 5, 1985 image (Landsat TM, path 233, rows 87–89). Note: Standard deviations are the square root of the eigenvalues. Rotation contains the eigenvectors

PC1	PC2	
2001 proportion data		
Tree plantations	−0.19	−0.92
Native Forest	0.91	−0.04
Ag/Ranch land	−0.36	0.39
Cleared land	−0.09	−0.03
S.D.	25.51	8.82
Eigenvalues	650.76	77.79
Cumulative% variance explained	70%	86%
1985 proportion data		
Tree plantations	−0.02	−0.21
Native forest	0.90	−0.40
Ag/Ranch land	−0.38	−0.59
Cleared land	−0.20	0.67
S.D.	26.85	4.10
Eigenvalues	720.92	16.81
Cumulative% variance explained	79%	96%

Appendix D. Tree plantations did not typically become established in areas that were identified as suitable for agriculture [classes II and III] (CIREN and CORFO, 1983). Tree-plantations are becoming established on the foothills since higher soil class numbers also indicates higher sloped land. Soil classes IV–VI are classified as ideal for plantations and class VII is not ideal for plantations



Appendix E. When tree plantation and native forest classes are aggregated in the canonical correlation analysis, the results indicate that tree cover is related with lower shellfish quality essentially masking the effect of tree plantations in the model where tree plantations are isolated. The standardized canonical coefficients (SCC) for canonical variables of landscape and seascape characteristics (LCLUC variables) are compared for isolated (only tree planation) (46) and aggregated (tree plantation and native forest) analyses. The standardized canonical coefficients (SCC) for the shellfish characteristics (for both original and aggregated analyses) both represent lower shellfish quality

Tree plantations isolated		Tree plantations & native forest aggregated	
	SCC		SCC
LCLUC Variables			
PAG_CH	0.187	PAG_CH	−0.235
PCL_CH	0.245	PCL_CH	0.295
PURB_CH	0.028	PURB_CH	0.014
PMAT_CH	0.036	PMAT_CH	0.089
PL_CH	0.294	TCOV_CH	0.612
PPL_CH	0.702	PTCOV_CH	0.135
CHLA	0.399	CHLA	0.566
Shellfish Variables			
WGHMT	0.253	WGHMT	0.201
SHELLGT	−0.188	SHELLGT	−0.077
SHELLHT	−0.069	SHELLHT	−0.120
PBARN	0.438	PBARN	0.459
PPOLYC	0.193	PPOLYC	0.178
PPHOR	0.675	PPHOR	0.672
PBIVAV	0.240	PBIVAV	0.141
PORLIP	−0.032	PORLIP	−0.031

Note: land cover change is from 1985 to 2001 and PAG_CH = percent change in agriculture, PCL_CH = percent cleared land change, PURB_CH = percent urban land class change, PMAT_CH = percent matorral change, PL_CH = area (km^2) plantation change, and PPL_CH = percent plantation change. CHLA = average chlorophyll a characteristics in the management area, TCOV_CH = total tree plantations and native forest change, and PTCOV_CH = percent tree plantations and native forest change. WGHMT = loco meat weight (g), SHELLGT = shell length (cm), SHELLHT = shell height (cm), PBARN = percent barnacles on shell, PPOLYC = percent polychaetes on shell, PPHOR = percent phoronids on shell, PBIVAV = percent boring bivalves on shell, PORLIP = percent lipid in loco meat.

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